1. DATA UNDERSTANDING

Data Understanding

The dataset contains demographic, financial, and contact history information for Banco Atlas customers. The objective is to identify which characteristics are associated with a higher probability of subscribing to a term deposit following a marketing campaign.

Missing values are found in the `age`, `marital`, and `education` variables. Since the percentage of missing data is very low, these records can be removed during the data cleaning phase without significantly affecting the dataset.

The target variable (`deposit`) is almost perfectly balanced between customers who subscribed to the term deposit and those who did not (approximately 49.7% vs. 50.3%).

The dataset population is composed mainly of middle-aged adults. The `balance` variable shows a strong positive skew and contains potential outliers, although these may represent customers with high account balances. The `duration` variable is also right-skewed, indicating that most phone calls are relatively short.

Dataset overview

- 10,642 records
- 18 variables
- 1 record = 1 contacted customer

Group	Variables
Identifier	`id`
Demographics	`age`, `job`, `marital`, `education`
Financial	`balance`, `default`, `housing`, `loan`
Campaign	`contact`, `day`, `month`, `duration`, `campaign`
History	`pdays`, `previous`, `poutcome`
Objective	`deposit`

Missing values (NaN)

- `age`: 9
- `marital`: 5
- `education`: 7

Descriptive statistics highlights

- `balance` → min −6,847 / max 81,204
- `campaign` → max 63 contacts performed during this campaign for a single client
- `duration` → max 64 minutes
- `pdays` → −1 means the client was never previously contacted

Unknown values

Variable	Unknown count
`poutcome`	7,885
`contact`	2,196
`education`	473
`job`	68

Key distributions

`poutcome` — failure: 1,180 · success: 1,065 · other: 512 · unknown: 7,885

`deposit` — yes / no ≈ 50% balanced

`contact` — cellular: 7,707 · unknown: 2,196 · telephone: 739

`month` — max: May (2,646 contacts) · min: December (109 contacts)

Variable reference table

Variable Name	Role	Type	Demographic	Description
age	Feature	Integer	Age	—
job	Feature	Categorical	Occupation	type of job (categorical: 'admin.', 'blue-collar', 'entrepreneur', 'housemaid', 'management', 'self-employed', 'services', 'student', 'technician', 'unemployed', 'unknown')
marital	Feature	Categorical	Marital Status	marital status (categorical: 'divorced', 'married', 'single', 'unknown'; note: 'divorced' includes widowed)
education	Feature	Categorical	Education Level	(categorical: 'basic.4y', 'basic.6y', 'basic.9y', 'high.school', 'illiterate', 'professional.course', 'university.degree', 'unknown')
default	Feature	Binary	—	has credit in default?
balance	Feature	Integer	—	average yearly balance
housing	Feature	Binary	—	has housing loan?
loan	Feature	Binary	—	has personal loan?
contact	Feature	Categorical	—	contact communication type (categorical: 'cellular', 'telephone')
day_of_week	Feature	Date	—	last contact day of the week
month	Feature	Date	—	last contact month of year (categorical: 'jan', 'feb', 'mar', …, 'nov', 'dec')

duration	Feature	Integer	—	last contact duration, in seconds. Important note: this attribute highly affects the target (e.g., if duration = 0 then deposit = 'no'). The duration is not known before a call is performed, and after the end of the call the outcome is obviously known. Thus, this input should only be included for benchmark purposes and should be discarded if the intention is to have a realistic predictive model.
campaign	Feature	Integer	—	number of contacts performed during this campaign and for this client
pdays	Feature	Integer	—	number of days that passed after the client was last contacted from a previous campaign (−1 means client was not previously contacted)
previous	Feature	Integer	—	number of contacts performed before this campaign and for this client
poutcome	Feature	Categorical	—	outcome of the previous marketing campaign (categorical: 'failure', 'nonexistent', 'success')
deposit	Target	Binary	—	has the client subscribed to a term deposit?

2 EDA

Exploratory Insights and Next Steps

The Exploratory Data Analysis (EDA) provided a comprehensive understanding of the banking dataset and revealed several patterns that will support the subsequent stages of the project.
The following insights summarize the most relevant observations obtained during this exploratory phase.

Data Quality Insights

- The dataset presents a high level of completeness, with only a limited number of missing values affecting a small subset of variables.
- Most variables exhibit consistent data quality and are suitable for subsequent analytical stages.
- Several categorical variables contain the category **"unknown"**, which should be carefully evaluated during the Data Cleaning phase to determine the most appropriate treatment.
- Numerical variables such as **balance**, **duration** and **campaign** display skewed distributions and the presence of extreme values. Based on the exploratory analysis, these observations appear plausible within a banking environment and should not automatically be considered data errors.

Customer Profile Insights

- Customer age shows only moderate differences between subscribers and non-subscribers, suggesting that age alone may not fully explain deposit subscription behaviour.
- Occupation appears to influence customer behaviour, with profiles such as retired customers and students showing relatively higher subscription rates compared with several other occupational groups.
- Customers with tertiary education present higher subscription proportions than customers with primary education, indicating that education level may contribute to future customer segmentation analyses.
- Subscription behaviour also varies across marital status categories, suggesting that demographic variables may become useful explanatory features during later business analyses.

Financial Insights

- Customers who subscribed to a term deposit generally present higher account balances than customers who did not subscribe.

- Customers with previous credit default records appear to show lower subscription rates, suggesting that financial variables deserve further investigation during the Financial & Credit Risk analysis.
- Exploratory comparisons between **balance**, **loan**, **housing** and **default** reveal behavioural patterns that justify deeper analysis in the subsequent business stage.
- The financial variables analysed in this section provide a strong foundation for answering the business question assigned to the Financial & Credit Risk Analytics team.

Campaign Insights

- Campaign-related variables reveal noticeable differences between customer groups and provide valuable context regarding customer interactions throughout the marketing campaign.
- Variables such as **duration**, **campaign**, **contact** and **month** describe operational characteristics of the campaign and may contribute to future business interpretation, although no marketing conclusions are drawn during the EDA stage.

Historical Campaign Insights

- Variables describing previous marketing campaigns (**previous**, **pdays** and **poutcome**) provide valuable historical context regarding customer interactions.
- Previous campaign outcomes appear to contain useful information that may contribute to understanding current customer behaviour and should therefore be considered during future analytical stages.

Correlation Insights

- Correlation analysis did not reveal strong linear relationships among the numerical variables included in the dataset.
- This suggests that customer behaviour is likely influenced by the interaction of multiple variables rather than by a single numerical factor.
- Since several important predictors are categorical, additional analytical techniques beyond Pearson correlation may be required during later stages of the project.

Overall Assessment

The Exploratory Data Analysis successfully achieved its primary objective of understanding the structure, characteristics and behaviour of the banking dataset.
The analyses performed throughout this notebook provide a solid analytical foundation for the remaining phases of the project while preserving the integrity of the original data.

No modifications, transformations or cleaning operations were applied during this stage, ensuring that the subsequent Data Cleaning process will be performed on the original dataset.

Next Steps

The insights obtained during this Exploratory Data Analysis will directly support the following stages of the project:
- 3. Data Cleaning
- 4. Data Transformation
- 5. Data Reduction
- Business Question Analysis
- KPI Definition
- Power BI Dashboard Development
- Business Recommendations

The exploratory findings presented in this notebook will also support the two analytical teams within **Team 32** by providing an initial understanding of both customer demographic profiles and customer financial behaviour prior to addressing the business questions assigned to each group.

3. Data Cleaning

Data Cleaning

The data cleaning process was designed to improve data quality while preserving as much relevant information as possible for the analysis.

The following steps were performed:

- **Duplicate check:** The dataset was inspected for duplicate rows and duplicate customer IDs to ensure that each observation represented a unique record.
- **Column review:** All variables were examined to identify irrelevant or redundant columns. Only variables considered useful for the analysis were retained.
- **Handling missing values:**
    - Records with missing values in the `marital` and `education` columns were removed, as these variables were considered essential for the analysis.
    - Missing values in the `age` column were imputed using the median, as it provides a robust estimate that is less sensitive to extreme values.
- **Outlier treatment:** Numerical variables such as `age`, `balance`, `duration`, `campaign`, `pdays`, and `previous` were analyzed to identify potential outliers. Instead of removing extreme observations, a conservative approach was adopted. For highly skewed variables, outliers were handled using **winsorization**, capping extreme values at predefined thresholds. This reduced the influence of unusually large or small values while preserving the overall distribution and avoiding unnecessary information loss.
- **Data type validation:** Variables were assigned appropriate data types:
    - Numerical features (`age`, `balance`, `day`, `duration`, `campaign`, `pdays`, and `previous`) were converted to numeric format.
    - Categorical features (`job`, `marital`, `education`, `default`, `housing`, `loan`, `contact`, `month`, `poutcome`, and `deposit`) were converted to string type to ensure consistency during exploratory analysis and feature engineering.
- **Feature engineering:** Two additional variables were created to capture previous customer interactions:
    - `no_previous_contact`: Indicates whether the customer had never been contacted before (`pdays = -1`).
    - `had_previous_contact`: Indicates whether the customer had at least one previous interaction (`previous > 0`).

These preprocessing steps resulted in a clean and consistent dataset, reducing issues related to missing values, inconsistent data types, duplicate records, and extreme observations while creating additional features that may improve subsequent analysis and predictive modeling.

4. Data transformation

Data Transformations

The data transformation process was designed to enrich the cleaned dataset with analytically meaningful variables. New columns were derived from existing features to support customer profiling, segmentation, and dashboard visualisation in Power BI. All transformations were applied after the cleaning stage and exported as a single shared CSV for use across all analysis notebooks.

The following transformations were performed:

- **Age group (`age_group`)**
  - The continuous variable `age` was discretised into five segments using manually defined breakpoints based on life-stage logic. The resulting categorical variable `age_group` provides a meaningful lens for comparing subscription behaviour across stages of the customer lifecycle.
  - Breakpoints: 18–25 (`Young`), 26–35 (`Young Adult`), 36–50 (`Adult`), 51–65 (`Middle-Aged`), 65+ (`Senior`). Intervals are right-closed. Nine records with missing age values were excluded.

- **Education encoding (`education_rank`, `education_label`)**
  - The ordinal variable `education` was encoded in two complementary forms. `education_rank` assigns a numeric score (Primary = 1, Secondary = 2, Tertiary = 3) for correlation analysis and future modelling. `education_label` provides a clean readable label for dashboard visualisation.
  - Records with `unknown` education were assigned `NaN` in the rank column and retained as a separate `Unknown` category in the label column, preserving 473 observations that would otherwise be lost.

- **Job profile (`job_profile`)**
  - The twelve original job categories in `job` were consolidated into six interpretable profiles to reduce dimensionality while preserving behavioural differences observed in the data.
  - Grouping logic: `admin.`, `management`, `technician` → `White Collar`; `blue-collar`, `housemaid`, `services` → `Blue Collar`; `entrepreneur`, `self-employed` → `Self-Employed`; `student` → `Student`; `retired` → `Retired`; `unemployed`, `unknown` → `Other`.
  - Students and retired customers were given individual profiles after empirical analysis confirmed substantially different subscription rates (76.0% and 68.3% respectively) compared to other inactive segments.

- **Financial burden score (`financial_burden`, `financial_burden_label`)**

- A composite score was derived from three binary variables — `default`, `housing`, and `loan` — each mapped to 1 (yes) or 0 (no/unknown). The resulting `financial_burden` column ranges from 0 to 3 and reflects the number of active financial commitments per customer.
- Values of `unknown` were treated as 0 (no burden assumed), a deliberate modelling choice that avoids penalising customers for incomplete data. A readable label column (`financial_burden_label`) maps scores to: `No burden`, `Low burden`, `Medium burden`, `High burden`.
- Intermediate scoring columns (`default_score`, `housing_score`, `loan_score`) were dropped before export.

- **Target variable encoding (`deposit`)**
  - The original binary target `deposit` (`yes`/`no`) was encoded as an integer (1/0) to align with Power BI DAX conventions and facilitate direct use in aggregation measures. This encoding was agreed across all team notebooks.

All transformed columns were appended to the cleaned dataset and exported as a shared CSV file (`bank_dataset_transformed_2026-07-01.csv`) to the `Data/` directory of the shared repository. This file serves as the single source of truth for all downstream analysis and dashboard development.

5. Data Reduction

Data Reduction

Do not proceed.

KPI's

KPIs — Power BI Dashboard

The following Key Performance Indicators were built in Power BI using DAX measures, grouped under a dedicated `_Mesures` table. The dataset was imported directly from the transformed CSV file, which serves as the single source of truth for all dashboard calculations.

All measures use `deposit = 1` as the subscription condition, consistent with the integer encoding agreed across team notebooks.

KPI 1 · Conversion Rate

Two measures were created in sequence. First, a count of customers with at least one active product (housing loan, personal loan, or term deposit). Second, a conversion rate expressing that count as a proportion of the total dataset.

```
Clients_amb_producte =
COUNTROWS(
    FILTER(
        BANK_marketing,
        BANK_marketing[housing] = "yes" ||
        BANK_marketing[loan]    = "yes" ||
        BANK_marketing[deposit] = "yes"
    )
)
```

```
Taxa_Conversio =
DIVIDE(
    COUNTROWS(
        FILTER(
            BANK_marketing,
            BANK_marketing[housing] = "yes" ||
            BANK_marketing[loan]    = "yes" ||
            BANK_marketing[deposit] = "yes"
        )
    ),
    COUNTROWS(BANK_marketing),
    0
)
```

KPI 2 · Average Call Duration (minutes)

This measure calculates the average duration of a contact call in minutes. The raw `duration` column is stored in seconds; the result is divided by 60 to convert to minutes.

```
Promig_duracio_trucades =
( SUM(BANK_marketing[duration]) / COUNT(BANK_marketing[id]) ) / 60
```

KPI 3 · Contact Channel Distribution (excluding unknown)

This measure calculates the percentage of contacts made via known channels (`cellular` or `telephone`), relative to all contacts where the channel is not `unknown`. This exclusion ensures the denominator reflects only records with valid contact information.

```
Porcentaje_Llamadas_Tel_Mobil =
DIVIDE(
    COUNTROWS(
        FILTER(
            BANK_marketing,
            BANK_marketing[contact] = "cellular" ||
            BANK_marketing[contact] = "telephone"
        )
    ),
    COUNTROWS(
        FILTER(
            BANK_marketing,
            BANK_marketing[contact] <> "unknown"
        )
    ),
    0
)
```

KPI 4 · Average Prior Contacts for Subscribers

This measure calculates the average number of campaign contacts made before a customer subscribed to the term deposit. Both the numerator and denominator are filtered to `deposit = 1` to restrict the calculation to confirmed subscribers only.

```
Promedio_Contactos_Previos_Suscriptors =
DIVIDE(
    CALCULATE(
        SUM(BANK_marketing[campaign]),
        BANK_marketing[deposit] = 1
    ),
    CALCULATE(
        COUNTROWS(BANK_marketing),
        BANK_marketing[deposit] = 1
    ),
    0
)
```

KPI 5 · Month with Highest Conversion Rate

Two companion measures were built for this KPI. The first returns the maximum conversion rate across all months. The second identifies the name of the month that achieved that rate. Both use `ADDCOLUMNS` to build an in-memory table of monthly conversion rates, with `VAR MesActual` used to capture row context correctly inside the iterator.

Month names were mapped to their full English equivalents during the transformation stage.

```
Conversion_Mes_Mayor =
VAR TaulaConversioMes =
    ADDCOLUMNS(
        VALUES(BANK_marketing[month]),
        "Conversio",
        VAR MesActual = BANK_marketing[month]
        RETURN
        DIVIDE(
            COUNTROWS(FILTER(BANK_marketing,
                BANK_marketing[deposit] = 1 &&
                BANK_marketing[month] = MesActual)),
            COUNTROWS(FILTER(BANK_marketing,
                BANK_marketing[month] = MesActual)),
            0
        )
    )
RETURN
    MAXX(TaulaConversioMes, [Conversio])
```

```
Max_mes_Tasa_Conversión =
VAR TaulaConversioMes =
    ADDCOLUMNS(
        VALUES(BANK_marketing[month]),
        "Conversio",
        VAR MesActual = BANK_marketing[month]
        RETURN
        DIVIDE(
            COUNTROWS(FILTER(BANK_marketing,
                BANK_marketing[deposit] = 1 &&
                BANK_marketing[month] = MesActual)),
            COUNTROWS(FILTER(BANK_marketing,
                BANK_marketing[month] = MesActual)),
            0
        )
    )
RETURN
    MAXX(TOPN(1, TaulaConversioMes, [Conversio]), BANK_marketing[month])
```

Extra · Filters for Further Analysis

Additional slicers were added to the dashboard to enable deeper segmentation in future analysis phases. These filters allow interactive exploration of KPI results across key demographic and campaign variables without requiring additional measures.

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Account Balance and Default Risk Analysis – Methodology and Technical Decisions

1. Objective

The objective of this analysis was to investigate whether a customer's account balance is associated with a higher probability of credit default and to identify customer segments with elevated default risk.

2. Variable Selection

The analysis focused on **account balance** as the primary explanatory variable and **default status** as the target variable. To further explore risk patterns, additional variables were incorporated, including **personal loan**, **housing loan**, **age group**, **job profile**, and a derived **financial burden index**.

3. Exploratory Analysis Strategy

The analysis followed an exploratory, descriptive approach. It began by examining the relationship between account balance and default using summary statistics and visualisations. Boxplots were used to compare balance distributions between defaulters and non-defaulters, while bar charts were used to compare default rates across customer segments.

This sequential approach allowed the analysis to progress from simple descriptive comparisons to a more detailed exploration of customer risk profiles.

4. Use of Default Rates

Customer groups differ substantially in size. Therefore, comparisons were based on **default rates (%)** rather than absolute numbers of defaults. Using percentages enables meaningful comparisons across segments regardless of the number of customers in each category.

5. Inclusion of Sample Size (n_customers)

Each reported default rate was accompanied by the corresponding number of customers. Providing the sample size improves the interpretation of the results, since identical default rates may represent groups with very different numbers of observations. This increases the transparency and reliability of the analysis.

6. Variable Transformation and Customer Segmentation

To improve interpretability and support business decision-making, several variables were transformed before the analysis.

- Account balance was segmented using both **quartiles** and **business-oriented balance ranges** to evaluate how default risk changes across financial situations.
- Age was grouped into broader categories to simplify comparisons.
- A **financial burden index** was created by combining customers' existing credit obligations, providing an overall measure of financial exposure.

These transformations reduce sparsity, improve interpretability, and facilitate the identification of meaningful customer segments.

7. Interaction Analysis

After analysing balance independently, interaction analyses were performed by combining balance with additional customer characteristics, including personal loans, housing loans, occupation, age group, and financial burden.

Grouped bar charts and summary tables were used to identify customer segments exhibiting higher default rates. This multivariate exploration provided deeper insights than analysing each variable separately and helped identify compound risk profiles.

8. Treatment of Small Groups

Customer segments with very small sample sizes were interpreted cautiously. Small groups can produce unstable default rates and potentially misleading conclusions. Although these observations were retained in the dataset, conclusions focused on customer segments with sufficient representation to ensure more reliable business insights.

9. Limitations

This study is based on **exploratory descriptive analysis**. Therefore:

- the results identify associations rather than causal relationships;
- no statistical hypothesis tests were performed to validate the observed differences;
- no predictive models were developed at this stage;
- the findings should be interpreted as exploratory evidence that can guide future modelling.

Future work should incorporate **inferential statistical tests** (e.g., Mann–Whitney U test or Chi-square tests) to verify whether the observed differences are statistically significant, followed by predictive modelling techniques such as logistic regression to quantify the impact of each variable on default probability.

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Customer Profile Analysis – Methodology and Technical Decisions

1. Objective

The objective of this analysis was to identify which demographic characteristics are associated with a higher probability of subscribing to a term deposit.

2. Variable Selection

The analysis focused exclusively on four demographic variables: Age, Job, Education and Marital status.

3. Exploratory Analysis Strategy

The analysis started with a univariate exploratory approach. Each demographic variable was analyzed independently to evaluate its relationship with the target variable (deposit subscription).

Bar charts were selected because they provide an intuitive comparison of subscription rates across categorical groups and facilitate the identification of general behavioural patterns before exploring more complex relationships.

4. Use of Subscription Rates

Customer groups differ considerably in size. For this reason, comparisons were based on subscription rates rather than absolute frequencies. Using percentages makes the results comparable across groups regardless of the number of observations contained in each category.

5. Inclusion of Sample Size (n_clients)

Each subscription rate was accompanied by the corresponding number of observations (n_clients). This provides additional context for interpreting the results, since identical percentages may represent groups with very different sample sizes. Including the sample size increases the transparency and robustness of the analysis.

6. Category Grouping

Several categorical variables were grouped into broader categories before the analysis. This decision was taken to reduce data sparsity, improve the interpretability of the results and minimize unstable estimates generated by categories with very few observations. Grouping categories produces more reliable and meaningful business insights.

7. Multivariate Analysis

After having analyzed each variable independently, pairs of demographic variables were combined using heatmaps. This approach allows interactions between demographic characteristics to be explored, revealing customer segments that may have remained hidden. The objective was to move from analyzing isolated variables to identifying complete customer profiles.

8. Treatment of Small Groups

Customer segments containing fewer than 30 observations were excluded from the interpretation of multivariate results. This because very small samples produce unstable percentage estimates and may generate misleading conclusions. These observations were not removed from the dataset but simply excluded from the interpretation of specific demographic combinations with insufficient statistical support.

9. Limitations

This study is based on descriptive statistical analysis. Therefore:

- the results identify associations rather than causal relationships;
- no statistical hypothesis tests were performed;
- no predictive models were developed;
- financial and behavioural variables were intentionally excluded from this analysis.

Future work could extend the analysis by incorporating inferential statistics and predictive modelling techniques to validate the observed demographic patterns.

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